



STRENGTHS

ISSUES

FOCAL PLAN

these are titles

GOALS

OBJECTIVES

FINAL BOOK

SUBTITLE

Header

Text



BRISTOL BOROUGH

EXISTING CONDITIONS

- **Demographics**
- **Economic vibes**
- **Housing**
- **Transportation**
- **Land use and zoning**
- **Environment**

EXISTING CONDITIONS ENVIRONMENT

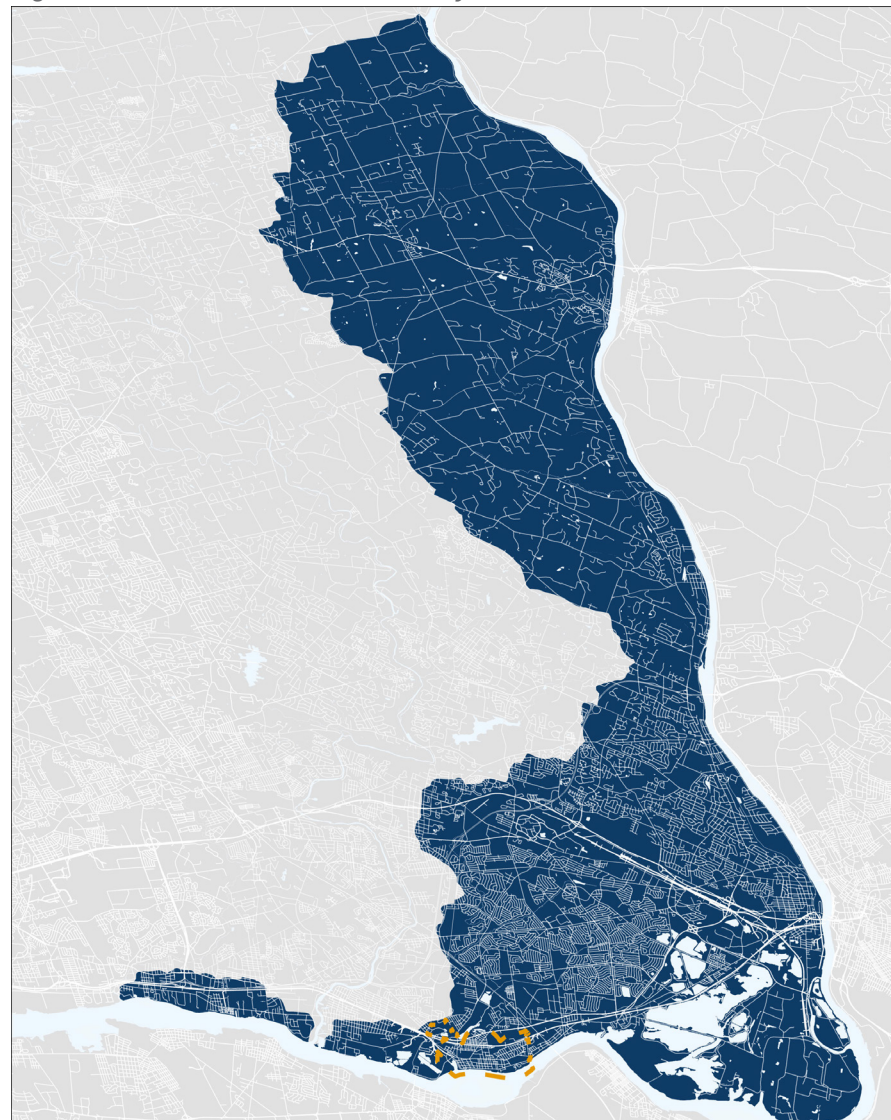
Bristol is a borough on the Delaware River, located 23 miles northeast of Philadelphia within the Delaware River South primary watershed. The watershed is home to 3 water treatment plants which serve over 1.6 million people within Bucks County and Philadelphia County. A watershed is an "area of land draining to a particular stream," according to the Delaware River Basin Commission, indicating how environmental disasters impact the entire watershed. The watershed which Bristol sits on is highlighted as environmental impacts aren't contained to political borders.

The borough is ecologically diverse with over 400 species of birds and over 100 species of fish calling the region their home. There are also three Natural Heritage Areas (NHA) located within the Borough: Bristol Marsh on its south-west, Silver Lake on its north, and the forest between the railroad spur and the canal on its north-east.

Figure X, Ecological Diversity i guess?



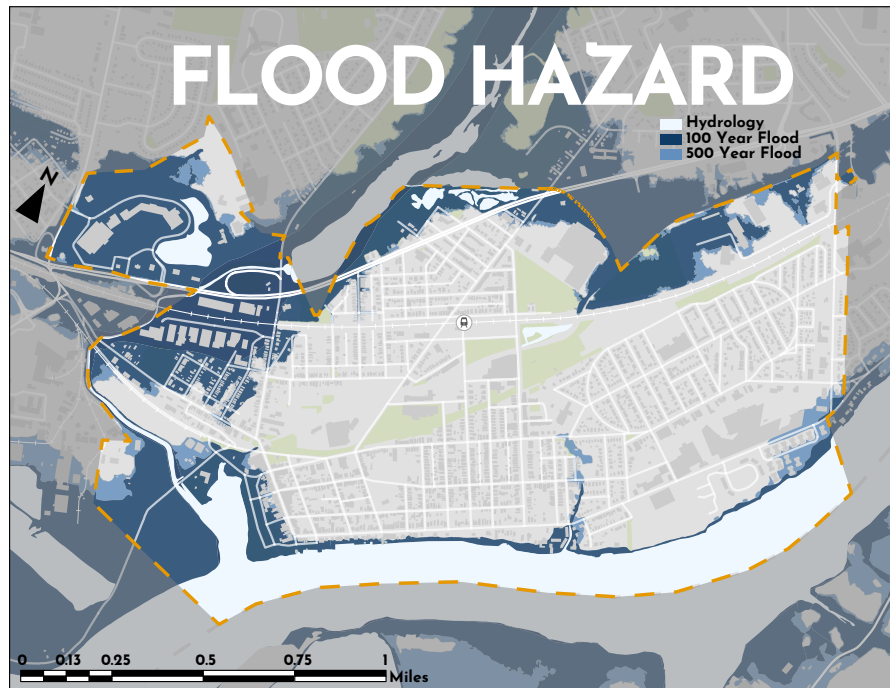
Figure X, Delaware River South Primary Watershed



EXISTING CONDITIONS ENVIRONMENT

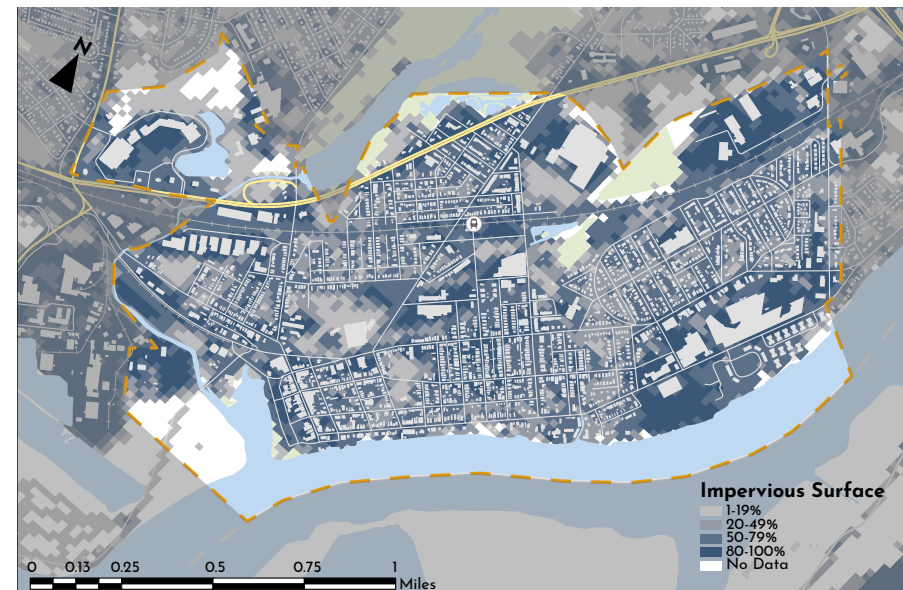
As a riverfront borough, sea level rise presents a critical threat to the community's future. Much of its waterfront properties and the southeastern corner lie within flood zones, leaving these areas acutely vulnerable. Compounded by climate change, the borough has experienced four major flooding events in the past four years: July 13, 2021; October 29, 2021; January 10, 2024; and April 4, 2024. Studies have shown Bristol Borough is facing a seawall construction cost of \$8,846,000 by 2040, making it the municipality with the 8th highest cost in Pennsylvania. These recurring disasters along with the increasing cost of seawall construction highlight the urgent need to address this escalating environmental challenge.

Figure X, Flood Hazard map



Another issue facing the borough is the presence of hazardous sites. There are 20 Hazardous Waste Sites (RCRA; Active), 4 brownfields properties, and 1 superfund site within the borough. A brownfield is land which has been developed for industrial purposes, polluted, then left abandoned. A superfund site is an abandoned hazardous waste site noted by the Environmental Protection Agency (EPA).

Figure X, Impervious Surfaces



With limited tree canopy coverage along with an increasing percentage of impervious surfaces within the borough, the urban heat island effect has shown itself to also be a major challenge facing the borough. Bristol has been shown to be within the 94th percentile within Bucks county for a municipality with the greatest benefit from opening a cooling center.



BRISTOL BOROUGH **ISSUES**

BRISTOL BOROUGH ISSUES



A couple issues are highlighted as the main overarching challenges facing the borough. This includes having underutilized lots, being a community that is largely car dependent, having a majority of the borough being reserved for low density housing, and the increasing threat of flooding and the hazards brought with that.

Under Utilized Lots

Many lots within the borough are vastly under utilized. This includes the waterfront lots by Mill St, and by Radcliffe St, along with the parking lot by Bristol Station. These parking lots take up large spaces providing very little use for the community. There are also many vacant lots around the Town Center which currently sit empty, contributing to the lack of vibrancy on Mill St. Many of these under utilized lots are municipally owned and can be leveraged to transform Bristol.

Figure X, Empty Lot



BRISTOL BOROUGH ISSUES

Car Dependency

While the core of Bristol is walkable, most amenities needed such as grocery stores are only accessible by car. Furthermore, with a lack of tree coverage in the neighborhood, a lack of amenities within walking distance, and free on street parking, residents are currently incentivised to remain car dependent.

Low Density Housing

Majority of the borough is covered by single family residential housing (70%). With the Harriman Historic District along with the presence of many historic buildings, this leaves Bristol with very little opportunity for growth.

Flood Hazard

Sitting by the Delaware River, the borough is susceptible to floodings. This is further exacerbated by climate change along with the western section of the borough being located within the flood hazard area. With over 4 major flooding events having occurred within the last 5 years, there is a great need for Bristol to adapt to the rising tides.

Figure X, Cars and big parking lot



Figure X, Housing type

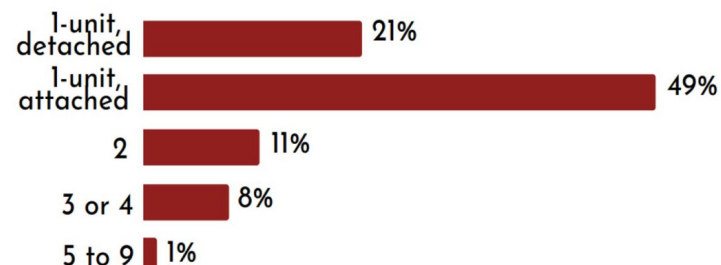


Figure X, Flooding





BRISTOL BOROUGH

OPPORTUNITIES

BRISTOL BOROUGH OPPORTUNITIES

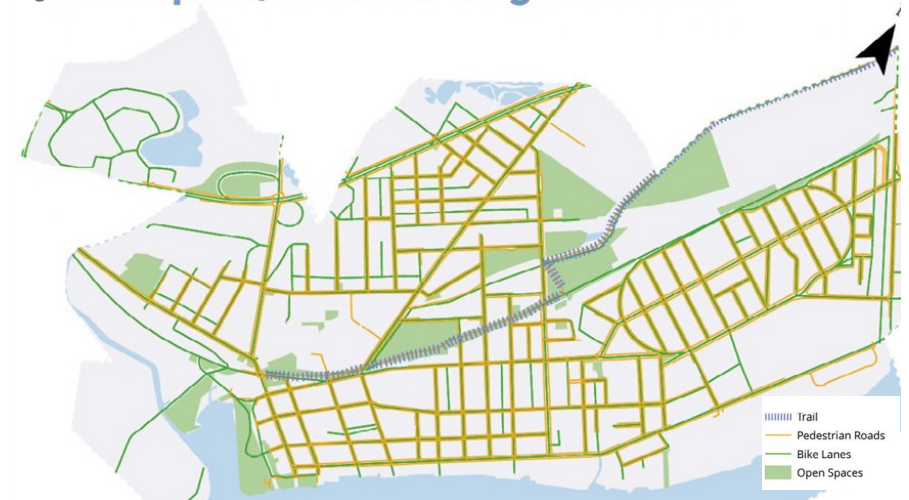


A couple opportunities are highlighted as areas the borough can leverage as it develops. This includes being located within an accessible location, the presence of historical buildings along with the boroughs historic roots, the Mill St. commercial corridor, and the overall walkability within the borough.

Walkability

With a walkable downtown core along with the Bristol Station within a 30-minute walk from most areas of the borough, Bristol is a walkable place for most people. Speaking with locals, they have called Bristol a “quiet, walkable neighborhood”, and that is reflected when looking at the pedestrian walkways.

Figure X, Businesses on Mill St.



BRISTOL BOROUGH OPPORTUNITIES

Commercial Corridor

The Mill St. commercial corridor serves as a centralised area where many businesses can thrive. Looking at the current businesses, there are 13 services, 16 food & beverage, 16 retail, and 2 cultural & entertainment.

Accessible Location

Located 38 minutes or 8 stops away from Philadelphia's 30th St Station, and 16 minutes or 2 stops away from the Trenton Transit Center, Bristol Station serves as an accessible location for commuters using the SEPTA Regional Rail Trenton Line.

Historical Buildings

The historic roots of the borough can be seen within its motto "Welcome Friend", which was on a wooden sign that greeted Marquis de Lafayette following the founding of the country. With 50 historical buildings along the waterfront alone, there is a distinct historic identity that the borough can leverage to increase tourism.

Figure X, Businesses on Mill St.



Figure X, Historic lots highlighted





THE PLAN

BRISTOL 2050

THE PLAN BRISTOL 2050

Introduction

Bristol 2050 is a visionary long-term plan, proposing bold strategies designed to transform the borough into a model of connectivity, resilience, equity, and vibrancy. Rooted in Bristol's unique identity and shaped by the needs of its residents, the plan builds on the borough's existing strengths—its accessible location, historic identity, its walkability, and commercial corridors—to reimagine what Bristol borough can become.

Plan Structure

Bristol 2050 follows a simple structure to ensure the plan is understandable for not only the policy makers, but also the residents of the borough. The overarching idea which drives the plan will be the vision statement. This statement represents what this plan imagines Bristol to be by 2050, with all other sections of the plan contributing to this vision. After the vision statement will be the four pillars. These four pillars can be better understood as the goals of the plan. Due to the vagueness of the words, further explanations will be given for what each pillar entails. Contributing to each pillar will be the objectives. Objectives are quantifiable metrics which will be used to ensure the plan is on track.

Bristol 2050 is structured around four core pillars: **Connected, Resilient, Equitable, and Vibrant**. These pillars, each guided by actionable objectives and strategies, are the driving force behind Bristol 2050. The following section will further explore what these pillar indicate, and how the objectives and strategies are connected.

Objectives will also be used in 2050 to see how successful the implementation of the plan was. Finally, the plan ends with the strategies, which are actionable recommendations to ensure the objectives, goals/pillars, and the vision of Bristol 2050 is met.



2050 Vision

“ By 2050, Bristol borough will thrive **in balance with ecological systems** while being a **vibrant** hub of economic activity and a center of local and regional **connectivity**. It will be a **resilient, affordable, and equitable community** for all. ”

THE PLAN PILLARS/GOALS

Bristol is Connected

A connected Bristol is a place where residents of all ages and abilities are comfortable walking or biking for their everyday needs. Where the pedestrian experience is prioritized to bridge residential neighborhoods and commercial districts. Where people are not bound to their homes due to a lack of curb cuts. By physically making Bristol connected, the borough will also become socially connected. A connected community where people care for one another and a community that works as one.

Bristol is Equitable

An equitable Bristol is a place where housing is a right, not a privilege, and where underutilized public spaces—like parking lots and vacant land—are transformed into assets that uplift the community. An equitable Bristol also means people of all ages and abilities are able and encouraged to go outside. Where residents of all income levels can find reprieve from a harsh summer's day, and where owning a car isn't a necessity of life.

Bristol is Resilient

A resilient Bristol is a place where communities thrive amid climate challenges, with adaptive infrastructure, green corridors to combat extreme heat, and flood-resistant landscapes that safeguard people and ecosystems. Through adaptive infrastructure, Bristol will become resilient in the way of being resilient to changes in the policy. A resilient Bristol is one where the residents' voices are heard and where the community can thrive despite changes in the climate or politics.

Bristol is Vibrant

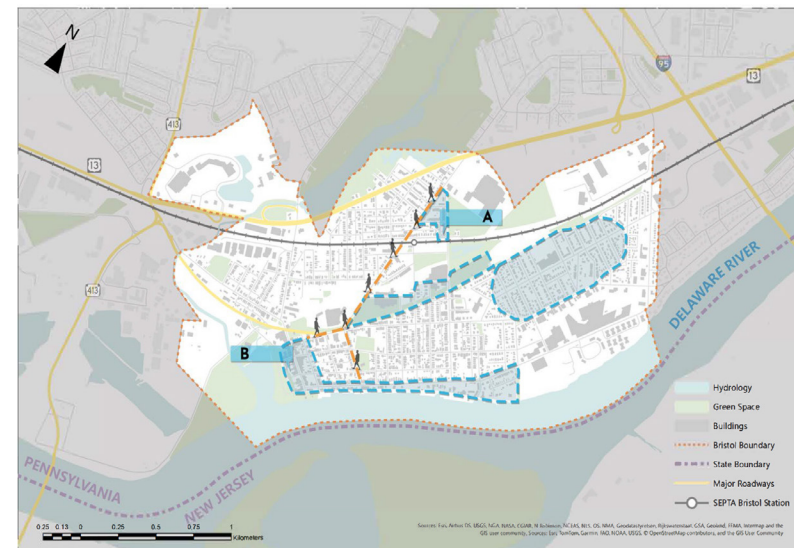
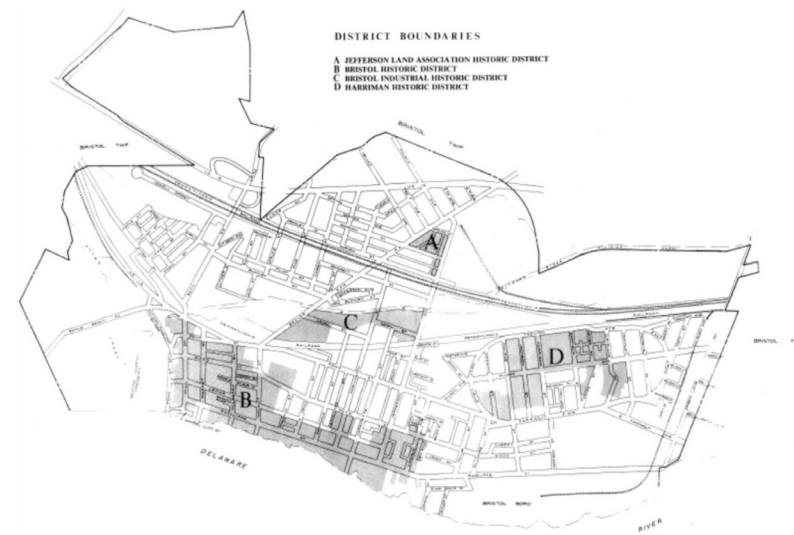
A vibrant Bristol is a place where small businesses are booming, residents are spending time together outside, and people are drawn in to visit from outside the borough—fueled by lively streetscapes, thriving third spaces, and a culture of creativity. This includes being able to connect to a variety of destinations, especially schools and transit. Neighborhoods that are pleasant and popular places to walk tend to be some of the city's most economically vibrant areas, and that improving pedestrian conditions can positively impact the liveliness of a neighborhood.

THE PLAN FOCAL AREA

Why these areas?

Following the district boundaries created by Bristol's planning commission, Bristol is split into four main historic districts noted as A (Jefferson Land Association Historic District), B (Bristol Historic District), C (Bristol Industrial Historic District), and D (Harriman Historic District). As the houses within the Harriman Historic District are protected, our plan will focus on the areas connecting A to B.

To ensure the plan isn't solely focused on one area, the connectivity between the districts will be prioritized. As seen in the picture on the bottom right, the main focus of the plan is the streets which connect people to and from Mill St and Bristol Station. This was decided upon due to the existing Spur line trail serving as a solid foundational path which currently connects residential areas to Mill St. By prioritizing the connectivity between the districts, we can ensure that changes to the town center area will benefit everyone living in the borough, rather than only benefiting the people who live by that area.



THE PLAN OBJECTIVES

1. Add 10 miles of protected bike lanes by 2050.

There are currently 27.31 miles of public roads in Bristol, with 21.14 miles maintained by the borough and the rest being maintained by the Pennsylvania Department of Transportation (PennDOT). While there are currently sharrows in the borough, there are no protected bike lanes other than the Spur line trail which is a multi-modal trail.

2. Install accessible pedestrian signals at 100% of signalized intersections by 2050.

This ensures that visually impaired residents can navigate streets safely and independently. Accessibility promotes equity and reduces pedestrian accidents.

3. Ensure 100% of sidewalks and curbs are ADA-compliant by 2050.

ADA-compliant infrastructure allows wheelchair users and people with mobility challenges to move freely. Removing physical barriers fosters equitable access to public spaces.

4. Reduce vehicular traffic volumes in the Town Centre by 40% by 2050

Fewer cars mean less congestion, lower emissions, and safer streets for pedestrians and cyclists. Prioritizing transit and walkability enhances quality of life and economic vibrancy.

5. Increase Bristol Station ridership by 30% by 2050

Indicates less reliance on cars and more people coming to Bristol.

6. Increase tree canopy coverage to 50% by 2050

More trees reduce urban heat, improve air quality, and support local wildlife. Native species require less maintenance and are better adapted to local conditions.

7. Achieve 80% capture of combined sewer overflow by 2050 through green stormwater infrastructure

Green infrastructure like bioswales and permeable pavement prevents flooding and pollution, protecting waterways

THE PLAN OBJECTIVES

8. Retrofit 30% of public right-of-way into nature-based assets by 2050

These green spaces absorb stormwater and mitigate climate impacts. It also reduces the road width.

9. Reduce urban heat island effect by 4°F by 2050

Bristol has one of the most heat vulnerable communities.

10. Ensure 100% of critical infrastructure is flood-proofed by 2050.

Protecting hospitals and emergency services ensures continuity during disasters. Flood barriers like AquaFence were used in Tampa General Hospital and proven to be necessary.

11. Convert 50% of underutilized lots into mixed-income housing, parks, or community gardens by 2050.

Transforming vacant land helps create third spaces while also cleaning up polluted lots.

12. Ensure 100% of public buildings (libraries, community centers) are fully ADA-compliant by 2050.

Public spaces should be accessible to the public. That includes people with disabilities.

13. Zone 100% of the Town Centre for mixed-use development by 2050

Mixed-use zoning creates walkable neighborhoods and vibrant street life which supports small businesses and reduces cars.

14. Create 10 new “third spaces” by 2050

There’s no third spaces right now. Third space is good.

15. Reallocate 100% of waterfront parking lots into pedestrian promenades, outdoor markets, and cultural event spaces by 2035.

THE PLAN STRATEGY

1. Build a 15 minute City Framework

A 15-minute city—where residents meet daily needs within a short walk or bike ride—requires radical zoning reforms and a prioritization of being pedestrian-first. Bristol should zone 100% of its town center, along with the corridor between the focal areas, as mixed-use development. Looking to *Paris* as an example, after zoning the entire city as mixed-use, car trips in the city were cut by 40%.

This can be furthered by Car-free zones around schools and transit hubs while building protected bike lanes which connect neighborhoods. An example of this would be to implement similar protected bike lanes seen in Philadelphia. Temporary car-free days, such as the Open Streets: West Walnut in Center City *Philadelphia*, can be done to build public support before implementing more permanent solutions.

Goal Alignment: *Connected, Equitable, Vibrant*

Cost: \$\$

Vibe: Very much vibes



THE PLAN STRATEGY

2. Prioritize Green Infrastructure and nature based solutions

With the increasing risk of flood vulnerability and heat vulnerability, there is a great need for increased green infrastructure within Bristol. This can be done by mandating features like bioswales, permeable pavement, and green roofs in new developments. An example of this would be *Portland's Grey to Green Initiative*, which saw reductions of sewer overflows by 40% due to the increased on-site stormwater management.

The prioritization of nature based solutions also entails the addition of urban forest corridors lining the streets, especially around neighborhoods with low-canopy coverage. An example of this would be the *Greenest City Action Plan in Vancouver* where specific regions with less tree canopy coverage were found, and native species were planted and maintained.

Finally, the under utilized lots can be transformed into multi-functional third spaces; Both as a gathering place, and as a flood control basin. An example of this would be *Singapore's ABC Water program*.

Goal Alignment: *Resilient, Connected, Equitable*

Cost: \$\$\$

Vibe: Very much vibes



THE PLAN STRATEGY

3. Launch “School Streets” Initiative

Reimagining the areas around Bristol’s schools as pedestrian-first spaces would lead to major benefits for children’s health, overall air quality, and pedestrian safety. An prime example of this would be the over 500 car-free school streets in *London* which block off streets by schools during pick up and drop off times. Bristol could implement timed road closures during drop-off periods, using removable barrier or volunteer monitors. Another example of this would be the school streets in *New York* which use metal barriers to block off streets by certain schools. To ensure accessibility, there should also be secure bike parking and ADA-compliant sidewalks.

By transitioning all schools to car-free zones by 2050, Bristol would create safer, quieter environments that encourage active transportation while reducing traffic congestion and traffic accidents across the borough.

Goal Alignment: *Connected, Equitable, Vibrant*

Cost: \$

Vibe: Much vibes



THE PLAN STRATEGY

4. Transform the Bristol into a “Sponge City”

To address increasing flood risks, Bristol should adopt comprehensive “sponge city” principles inspired by Chinese cities like *Shenzhen*. This goes beyond traditional green infrastructure by designing the entire urban landscape to absorb, store, and reuse rainwater. Bristol can transform the parking lot by Bristol Marsh, along with the area around the Delaware Canal Park into a floodable park.

This would help reduce the increasing costs due to flooding, while also creating more third spaces. Furthermore, the creation of “sponge cities” would lead to increased biodiversity which will lead to a more vibrant and attractive outdoor environment. Finally, the creation of sponge cities also helps with reducing the urban heat island effect by increasing the blue-green corridors throughout the borough. Steps would include daylighting buried streams such as Otter creek.

Goal Alignment: *Resilient, Connected, Equitable*

Cost: \$

Vibe: Very very very much vibes

